

Estimation Theory - Exercise 3

- Submission date: 10/6/26 at 23:59 via the Moodle website.
 - Keep a clear and organized presentation (handwritten or typed).
 - Unexplained results will not be accepted.
 - Please write your name and i.d. or passport number on the first page.
1. Consider the observation model $x_n = \log \theta + w_n$ $n = 1, \dots, N$, where $w_n \sim N(0, 1)$ are i.i.d. and $\theta > 0$ is an unknown deterministic parameter.
- a. Propose an estimator of θ from the observations $\mathbf{x}^{(N)} = [x_1, \dots, x_N]^T$.
 - b. Using MATLAB/ Python, evaluate the bias, variance, and MSE of the estimates for different values of N and θ as follows. Let $\hat{\theta}_N = \hat{\theta}(\mathbf{x}^{(N)})$ denote the proposed estimator and define the estimation error for the i^{th} realization of $\mathbf{x}^{(N)}$ as $\varepsilon_N(\mathbf{x}_i^{(N)}, \theta) = \hat{\theta}(\mathbf{x}_i^{(N)}) - \theta$ where $i = 1, \dots, L$ stands for an index of the i^{th} realization of $\mathbf{x}^{(N)}$. The bias, variance, and MSE of the estimates are given by

$$\text{bias}(N, \theta) = \frac{1}{L} \sum_{i=1}^L \varepsilon_N(\mathbf{x}_i^{(N)}, \theta), \quad \text{var}(N, \theta) = \frac{1}{L} \sum_{i=1}^L (\varepsilon_N(\mathbf{x}_i^{(N)}, \theta) - \text{bias}(N, \theta))^2$$

$$\text{MSE}(N, \theta) = \frac{1}{L} \sum_{i=1}^L \varepsilon_N^2(\mathbf{x}_i^{(N)}, \theta),$$
 respectively, and $L = 1000$.
 - i. Plot the bias, variance, MSE of the estimator and the Cramér-Rao bound (CRB) for estimation of θ , versus N for at least 5 different values of N . In each figure, include plots for 6 different values of θ in the interval $0.1 \leq \theta \leq 10$ where the N axis is logarithmic. In the figures of the variance, and MSE use a logarithmic scale for both axes.
 - ii. Plot the bias, variance, and MSE of the estimator versus θ for at least 10 different values of θ in the interval $0.1 \leq \theta \leq 10$. In each figure include plots for 3 different values of N .
- Note:** Choose proper values of N to demonstrate your results. Attach the MATLAB/Python code and figures to your submitted work.
- c. Is the proposed estimator efficient?
 - d. Is the proposed estimator asymptotically efficient? Your answer should be based both on simulation results and on theoretical analysis.

2. Let $x_1, \dots, x_N$ be statistically independent Gaussian random variables with mean μ and standard deviation σ .
- Find the CRB for estimation $[\mu, \sigma^2]^T$. Is there any coupling between the two parameters in the Fisher information matrix (FIM)?
 - Does there exist an efficient estimator of $[\mu, \sigma^2]^T$? If yes, find the estimator.
 - Let $\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$, $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$. Is $\hat{\sigma}^2$ an unbiased estimator of σ^2 ? If it is not, propose a proper correction of this estimator to obtain an unbiased estimator of σ^2 that is μ -independent.
 - What is the variance of the estimator you obtained after the correction of $\hat{\sigma}^2$? Does this estimator achieve the corresponding CRB from Section a?
- Note:** You can use the fact that $\{x_i - \bar{x}\}_{i=1}^N$ are jointly Gaussian random variables with zero mean. In addition, you can use the following identity for zero-mean jointly Gaussian random variables:
- $$E[XYZW] = E[XY]E[ZW] + E[XZ]E[YW] + E[XW]E[YZ].$$
- Repeat Section d for $N \rightarrow \infty$.

3. Bhattacharyya bound:

- Prove that for any unbiased estimator of θ (denoted by $\hat{\theta}$), the following equality is satisfied:
$$E\left\{\left(\hat{\theta} - \theta\right) \frac{1}{f(\mathbf{x}; \theta)} \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i}\right\} = \begin{cases} 1 & i = 1 \\ 0 & i \geq 2 \end{cases}.$$
- Use the Cauchy-Schwarz inequality to find a lower bound for $E\left[(\hat{\theta} - \theta)^2\right]$.

Express the bound by the $n \times n$ matrix $\mathbf{Q}$, whose elements are given by

$$Q_{ij} = E\left\{\frac{1}{f^2(\mathbf{x}; \theta)} \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} \frac{\partial^j f(\mathbf{x}; \theta)}{\partial \theta^j}\right\}, \quad i, j = 1, \dots, n.$$

- Prove that the resulting bound in Section b is larger than or equal to the Cramér-Rao bound (Hint: Check the case $n = 1$).

Note 1: Assume that the matrix $\mathbf{Q}$ is invertible and that the partial derivatives, up to order n , of $f(\mathbf{x}; \theta)$ with respect to θ , exist and are absolutely integrable.

Note 2: Assume that the regularity condition of Cramér-Rao bound is satisfied.

4. The Cramér-Rao bound can be derived as follows. Define the vector

$$\mathbf{z} = \begin{pmatrix} \hat{\theta} - \theta \\ \frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta} \end{pmatrix}$$

- Verify that for any unbiased estimator $\hat{\theta}$, $E(\mathbf{z}) = \mathbf{0}$.
- Derive an expression for the covariance matrix of $\mathbf{z}$, $\mathbf{\Lambda} = E\{\mathbf{z}\mathbf{z}^T\}$.
- Use the fact that $\mathbf{\Lambda}$ is a positive semidefinite matrix to obtain the Cramér-Rao inequality.

Note: Assume that the regularity condition of Cramér-Rao bound is satisfied.

5. Let $w_1, \dots, w_N$ be i.i.d. random variables with probability density function (PDF)

$$f_{w_n}(w) = \frac{1}{2} \exp(-|w|). \text{ We are interested to estimate } \theta \text{ from the observations}$$

$$x_n = \theta + w_n, \quad n = 1, \dots, N.$$

- Confirm that the regularity condition of Cramér-Rao bound is satisfied.
- Compute the Cramér-Rao bound for estimation of θ from $x_1, \dots, x_N$.
- Does there exist an efficient estimator of θ ? If yes, find this estimator?

6. Let $\mathbf{x}$ be an $N \times 1$ Gaussian random vector with zero mean and covariance matrix $\mathbf{R}_x = \sigma^2 \mathbf{I}_N$, where σ^2 is an unknown deterministic parameter. Prove that an efficient estimator for σ^2 exists and find it.

- Prove that an efficient estimator for σ^2 , $\hat{\sigma}_{\text{eff}}^2$, exists and find it.
- Find the bias, variance, and MSE of $\hat{\sigma}_a^2 \triangleq a \hat{\sigma}_{\text{eff}}^2$ as a function of a and σ^2 .
- Find the optimal value of a , a_{opt} , that minimizes the MSE of $\hat{\sigma}_a^2$. Does a_{opt} depend on σ^2 ? If it does not, what is $\hat{\sigma}_{a_{\text{opt}}}^2$? If it does, what is $\hat{\sigma}_{a_{\text{opt}}}^2$ for $\sigma^2 = 1$?
- By using MATLAB/Python, compute and plot the bias of $\hat{\sigma}_{a_{\text{opt}}}^2$ from Section c and the bias of $\hat{\sigma}_{\text{eff}}^2$ on the same figure versus N (assume $\sigma^2 = 1$).
- By using MATLAB/Python, compute and plot the MSE of $\hat{\sigma}_{a_{\text{opt}}}^2$ from Section c and the MSE of $\hat{\sigma}_{\text{eff}}^2$ on the same figure versus N (assume $\sigma^2 = 1$).

7. We wish to estimate the variance of a zero-mean Gaussian random variable. The observations vector is Gaussian distributed: $\mathbf{x} \sim N(\mathbf{0}, \theta \mathbf{I}_N)$. The prior distribution of θ is Beta, with probability density function (PDF):

$$f_{\theta}(\theta) = \frac{\theta^{a-1} (1-\theta)^{b-1}}{\beta(a,b)}, \quad \beta(a,b) = \int_0^1 \nu^{a-1} (1-\nu)^{b-1} d\nu = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \quad \text{and}$$

$$\Gamma(a) = \int_0^{\infty} \nu^{a-1} e^{-\nu} d\nu \text{ is the Gamma function.}$$

- Calculate the conditional FIM.
- Calculate the prior FIM.
- Calculate the BCRB.